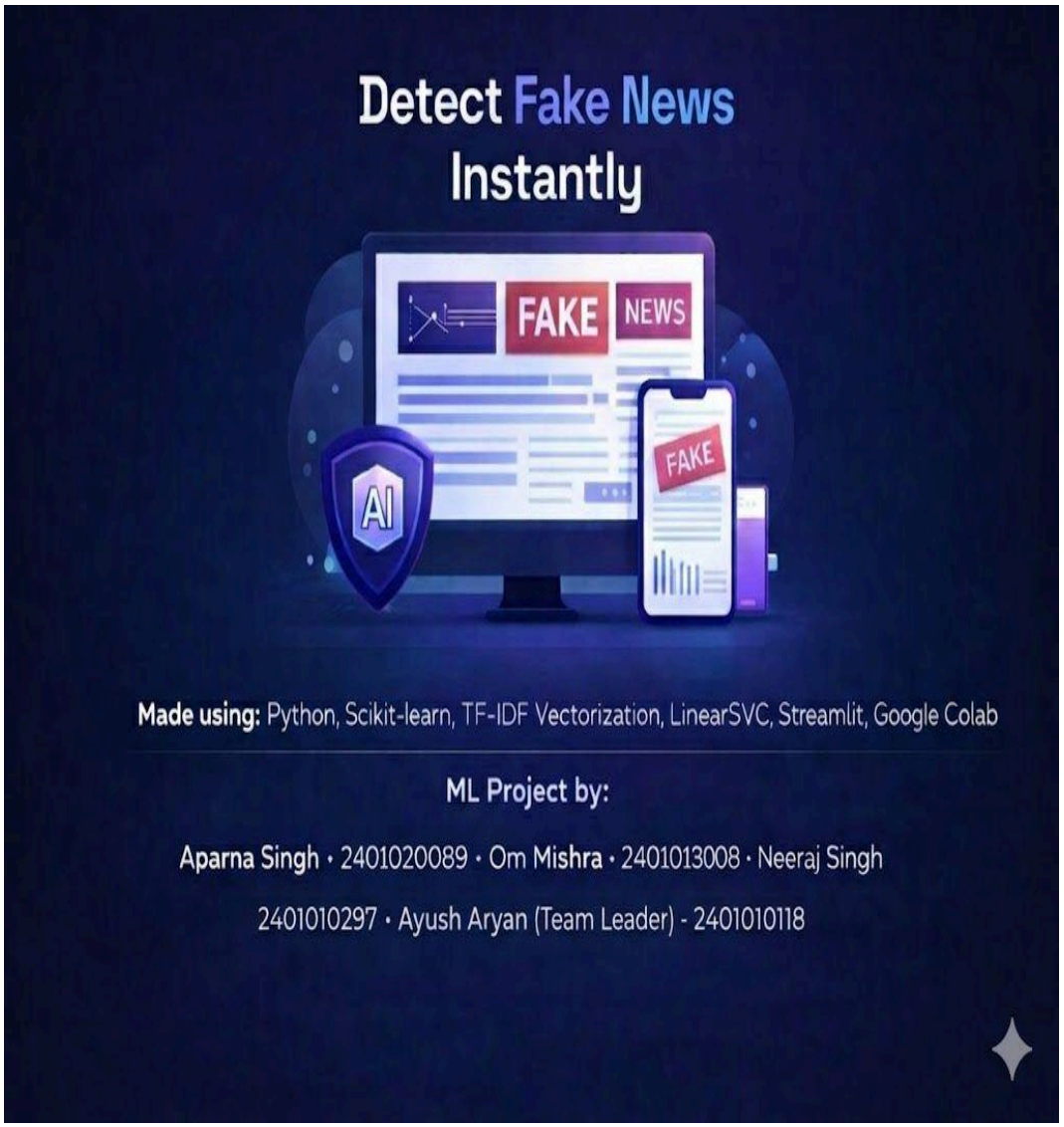
📘 News Credibility Detection & Writing Style Analysis System

📊 **Project Analysis Report**



1. Introduction

The rapid spread of misinformation in digital media demands automated systems that can evaluate news credibility in an interpretable and reliable manner. Most existing fake news detection models focus only on binary classification and behave as black-box predictors, offering little explanation of *why* an article is considered fake or real.

This project addresses this gap by developing a hybrid news credibility analysis system that combines statistical text classification with behavioural writing-style analysis.

🔹 **Project Vision (Overall Milestones)**

The complete project is designed in two stages:

- **Milestone 1 (Current Work):**
  Machine learning–based fake news classification with interpretable writing-style metrics.

- **Milestone 2 (Planned):**
  Extended credibility analysis framework with structured reasoning and advanced credibility assessment.

🔹 **What Has Been Implemented So Far**

In the current milestone, the core predictive and interpretability components were developed:

- Articles represented using **TF-IDF textual features**
- Classification using **Logistic Regression**
- Writing-style credibility metrics:
  - Sensational language
  - Clickbait framing
  - Exaggeration patterns
  - Visual emphasis

- Emotional intensity

These behavioural signals provide interpretable evidence alongside model predictions.

“Fake news is not defined only by incorrect facts; it often relies on emotional manipulation, sensational framing, and exaggerated claims. **By explicitly modeling these patterns, the system moves beyond simple classification toward transparent credibility analysis.”**

2 .Modeling Approach Selection

Before finalizing the fake news detection methodology, **multiple potential approaches were explored** to identify the most reliable and interpretable solution for credibility assessment.

Initially, the project considered a **multi-signal credibility modeling framework** combining **linguistic writing patterns, semantic consistency measures, and source reliability indicators**. This approach required **extracting full article text from URLs** and constructing **three feature matrices** representing stylistic manipulation, content coherence, and publisher trustworthiness. The objective was to develop a **richer credibility model beyond traditional text classification**.

However, during implementation, **large-scale article extraction introduced major challenges**. Many URLs failed to provide complete content, while others produced **noisy or partial text** due to website restrictions and formatting inconsistencies. This resulted in **uneven data quality** and **bias toward easily extractable sources**. Because reliable article text is essential for fair feature computation, the matrix-based approach risked **systematic distortion of model learning**.

A secondary idea explored **merging multiple extracted news datasets** to expand coverage before applying the matrix framework. Although this increased data volume, **cross-source inconsistencies and extraction**

reliability issues again affected dataset integrity. Consequently, this combined dataset was **not used for final modeling**.

After evaluating these limitations, the project adopted a **cleaner and more controlled pipeline based on TF-IDF text representation and logistic regression classification**, supplemented with **interpretable writing-style metrics**. This approach ensured **consistent input quality** while still providing behavioural credibility insights.

Final Decision: The selected design balances **predictive performance, interpretability, and dataset reliability**, making it suitable for robust fake news credibility analysis.

3. System Objectives

The project aims to:

- **Develop a supervised machine learning model** for reliable fake news detection
- **Prevent data leakage and artificial accuracy inflation** through careful preprocessing
- **Provide interpretable credibility scoring** rather than opaque predictions
- **Analyze writing style using behavioural linguistic metrics** associated with misinformation
- **Generate a structured analytical report** for any input news article

Overall Focus:
The system emphasizes not only classification accuracy but also **meaningful credibility evaluation and interpretable insight generation**.

4. Technical Architecture

The News Credibility Detection & Writing Style Analysis System follows a **modular pipeline architecture**, where each component performs a well-defined task in the overall processing workflow.

The architecture is designed to ensure:

- Clear separation of responsibilities
- Scalability and extensibility
- Interpretability of outputs
- Ease of model replacement or upgrade

🔄 System Workflow

Raw Article
↓
Text Cleaning
↓
TF-IDF Vectorization
↓
Logistic Regression Model
↓
Credibility Score (0–100)
↓
Writing Style Analysis
↓
Final Analytical Report

- **Detailed System Workflow**

  Step 1: Raw Article Input

The system accepts a news article consisting of:

- Title
- Main content text

The raw input may contain:

- Source identifiers
- Formatting artifacts
- Punctuation noise
- Dates and metadata

At this stage, the text is unstructured and unsuitable for direct machine learning processing.

Step 2: Text Cleaning & Preprocessing

The raw article undergoes systematic preprocessing to remove noise and prevent bias.

Operations performed include:

- Lowercasing all text
- Removing publisher indicators (e.g., “(Reuters)”)
- Eliminating day/month references
- Removing punctuation and numbers
- Removing predefined leakage keywords
- Normalizing whitespace

Architectural Purpose:

This stage ensures:

- Reduced dimensionality
- Prevention of shortcut learning
- Improved generalization
- Cleaner feature extraction

This component acts as a data integrity filter.

Step 3: TF-IDF Vectorization

After cleaning, the textual data is transformed into numerical form using Term Frequency–Inverse Document Frequency (TF-IDF).

This stage:

- Converts text into high-dimensional vectors
- Captures importance of words relative to corpus
- Includes unigrams and bigrams
- Limits vocabulary size to 20,000 features

Why This Stage Is Critical:

Machine learning models cannot process raw text directly.
TF-IDF creates a structured numerical representation where:

- Frequently occurring but less informative words are down-weighted
- Rare but discriminative words receive higher importance

This stage transforms qualitative text into quantitative features.

Step 4: Logistic Regression Model

The TF-IDF feature matrix is passed to a trained Logistic Regression classifier.

The model:

- Computes weighted linear combinations of features
- Applies sigmoid function
- Outputs probability of article being “Real”

Output:

A probability value between 0 and 1.

This probability reflects how strongly the article aligns with learned credibility patterns.

Step 5: Credibility Score Generation (0–100)

The predicted probability is scaled:

Credibility Score = Probability × 100

This produces a more interpretable metric for end users.

Instead of binary output:

0 → Fake
1 → Real

Users receive:

Example: 82% Credible

Architectural Significance:

This improves:

- User interpretability
- Ranking capability
- Threshold customization
- Practical deployment usability

Step 6: Writing Style Analysis Module

Parallel to ML prediction, the article is evaluated using rule-based behavioural metrics.

The system computes:

- Sensational Language Score
- Clickbait Framing Score
- Exaggeration Score
- Visual Emphasis Score

- Emotional Intensity (VADER Sentiment)

This module analyzes linguistic signals commonly associated with misinformation patterns.

Architectural Role:

This acts as a secondary analytical layer that:

- Enhances transparency
- Provides contextual explanation
- Supports interpretability
- Reduces black-box perception

Step 7: Final Analytical Report Generation

The final stage combines:

- Model prediction
- Credibility score
- Style metrics

Into a structured report containing:

- Predicted label
- Credibility percentage
- Linguistic indicators
- Behavioural signal analysis

This transforms the system from a classifier into a comprehensive credibility assessment tool.

5. Dataset & Preparation Analysis

Dataset Used

- True.csv → Label 1 (Real)
- Fake.csv → Label 0 (Fake)

Key Preparation Steps

- Dataset merging
- Label assignment
- Stratified train-test split (80/20)
- Title + Text combination
- Class distribution verification

🔎 **Analytical Observation**

The dataset contains strong publisher artifacts (e.g., Reuters patterns). The cleaning pipeline intentionally removes:

- “(Reuters)” tags
- Day/month references
- Publisher-related identifiers

6. Text Preprocessing & Leakage Prevention

The `clean_text()` function performs:

- Lowercasing
- Regex cleaning
- Punctuation & number removal
- Removal of bias-inducing keywords
- Whitespace normalization

7. Feature Engineering Evaluation

TF-IDF Configuration

- max_features = 20,000
- ngram_range = (1,2)
- min_df = 5
- max_df = 0.9
- stop_words = English

8. Model Selection & Justification

Model Used:

Logistic Regression

Strengths:

- Fast training
- Strong performance on sparse TF-IDF
- Interpretable coefficients
- Stable convergence

9. Model Evaluation Strategy

The system evaluates using:

- Accuracy
- ROC-AUC
- Confusion Matrix

🔎 Important Insight

The project correctly identifies:

- **False Positives** (Fake predicted as Real) → High risk
- **False Negatives** (Real predicted as Fake) → Credibility risk

10. Writing Style Analysis Module

This is the most innovative component of the system.

Unlike ML predictions, these are rule-based behavioural indicators.

Metric	**Purpose**
Sensational Score	Detects dramatic emotional words
Clickbait Score	Detects viral framing patterns
Exaggeration Score	Detects extreme wording

Visual Emphasis Score	Detects CAPS, !, ?
Emotional Intensity	VADER sentiment magnitude

11. System Strengths

- Strong baseline model
- Leakage prevention awareness
- Multi-layer evaluation
- Behavioral analysis integration
- Interpretable credibility score
- Clean architecture
- Proper train-test split

12. System Limitations

While the proposed system demonstrates effective credibility analysis, several limitations should be acknowledged:

- **Dataset Bias:**
  The dataset is heavily influenced by specific publishers (e.g., Reuters), which may cause the model to learn source-related patterns rather than fully generalizable misinformation characteristics.

- **No Cross-Validation:**
  Model performance was evaluated using a single train–test split, which may not fully capture variability across different data partitions.

- **Limited Semantic Understanding:**
  Logistic regression with TF-IDF features captures word frequency patterns but does not model deep contextual meaning or nuanced semantics present in language.

- **Rule-Based Style Metrics:**
  Writing-style indicators rely on predefined lexicons and patterns,

which may oversimplify or overgeneralize complex linguistic behaviour.

- **No Domain Generalization Testing:**
  The system was not evaluated on external or unseen news datasets, so real-world generalization across sources and topics remains unverified.

13. Improvement Recommendations

To further enhance the system and move toward industry-level robustness and deployment readiness, the following improvements are recommended:

- **Introduce K-Fold Cross-Validation:**
  Improve reliability of performance estimates and reduce dependence on a single train-test split.

- **Evaluate Linear SVM:**
  Test a strong text-classification baseline that often performs well with TF-IDF features.

- **Implement Gradient-Boosting Models (e.g., XGBoost):**
  Capture nonlinear feature interactions and potentially improve predictive accuracy.

- **Integrate Transformer-Based Models (e.g., BERT):**
  Incorporate contextual semantic understanding beyond bag-of-words representations.

- **Apply Probability Calibration:**
  Improve credibility score reliability so predicted probabilities better reflect true likelihood.

- **Use SHAP for Feature Interpretability:**
  Provide local and global explanations of model decisions for

transparency.

- **Develop a Streamlit Web Interface:**
  Enable interactive article analysis and practical usability.

- **Test on Out-of-Distribution Datasets:**
  Evaluate robustness on unseen news sources to assess real-world generalization.

14. Conclusion

The **News Credibility Detection & Writing Style Analysis System** provides a structured and interpretable approach to fake news detection by combining machine learning classification with behavioural linguistic analysis. The project demonstrates careful data preprocessing, effective feature engineering, and the practical application of supervised learning for credibility prediction.

Beyond predictive accuracy, the system focuses on interpretability by incorporating writing-style indicators such as sensational language, exaggeration, emotional intensity, and visual emphasis. These behavioural signals allow credibility assessment to move beyond simple binary classification toward meaningful analytical insight.

The evaluation process further reflects an understanding of model reliability through appropriate performance metrics and error interpretation. Overall, the project demonstrates the design of an applied NLP credibility analysis system that balances predictive modeling, behavioural signal integration, and interpretability, resulting in a transparent and analytically grounded approach to misinformation detection.